

Addendum to Version 1.0

The Unique AQUAL Interpolation Function for the Additive Modified Gravity Formula

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Four minor clarifications and additions following peer review.

Revision 1: Distinction from the McGaugh et al. (2016) RAR Fitting Function

The additive formula used in this paper, $g = g_{\text{bar}} + \sqrt{g_{\text{bar}} * a_0}$ [Eq. 1], should be clearly distinguished from the Radial Acceleration Relation (RAR) fitting function of McGaugh, Lelli & Schombert (2016), which takes the form:

$$g_{\text{RAR}} = g_{\text{bar}} * \nu(g_{\text{bar}} / a_0)$$
$$\nu(y) = 1/2 + \sqrt{1/4 + 1/y}$$

Expanding: $g_{\text{RAR}} = g_{\text{bar}}/2 + \sqrt{g_{\text{bar}}^2/4 + g_{\text{bar}} * a_0}$. The additive formula gives $g = g_{\text{bar}} + \sqrt{g_{\text{bar}} * a_0}$. These differ by the presence of the $g_{\text{bar}}^2/4$ term inside the square root of the RAR function. The quantitative discrepancy between the two formulas is shown in the table below.

Table A1. Comparison of the additive formula and the McGaugh et al. (2016) RAR function.

$f = g_{\text{bar}}/a_0$	g/a_0 (additive)	g/a_0 (RAR)	Difference (%)
0.01	0.110	0.110	0.2%
0.1	0.416	0.414	0.6%
0.5	1.207	1.186	1.8%
1.0	2.000	1.914	4.5%
2.0	3.414	3.236	5.5%
10.0	13.162	12.702	3.6%
100.0	110.000	109.513	0.4%

The two formulas agree to better than 1% in both the deep-MOND limit ($f \ll 1$) and the Newtonian limit ($f \gg 1$), but differ by up to 5.5% near $f = 2$ ($g_{\text{bar}} = 2*a_0$). Both are consistent with the SPARC data given observational uncertainties, but they are not identical. The interpolation function derived in this paper generates the additive formula exactly; it does not generate the RAR function. Deriving the AQUAL μ for the RAR function would require a separate calculation.

Revision 2: Prior Independent Derivation

After completion of this work, we became aware of an equivalent algebraic result posted on the Physics Forums website in June 2007 by an anonymous contributor (thread: 'Can MOND be ruled out at laboratory scale?', post #3). That post derives $\mu(x) = (\sqrt{1 + 1/4x} - \sqrt{1/4x})^2$, which is algebraically identical to our Equation 11. The forum post was not published in a peer-reviewed journal and was not accompanied by the Lagrangian free function $F(s)$, the stability analysis, the quantitative comparison with standard interpolation functions, or the uniqueness proof presented here. We acknowledge this independent prior derivation of the algebraic form.

Revision 3: Abstract Language Correction

The abstract of Version 1.0 states that the additive formula 'provides excellent fits to galactic rotation curves.' This should read: 'has been shown by independent studies (Verlinde 2017; Hossenfelder 2017; direct fits to SPARC data) to provide good fits to galactic rotation curves.' The present paper derives the Lagrangian foundation for this formula but does not itself perform rotation curve fits.

Revision 4: Scope Clarification in Conclusion

The conclusion of Version 1.0 states: 'This work provides the Lagrangian foundation for the additive class of modified gravity theories.' For clarity, this should read: 'This work provides the non-relativistic Lagrangian foundation for the additive class of modified gravity theories. The relativistic (Lorentz-covariant) extension, required for gravitational lensing predictions and cosmological applications, involves a distinct free function $F_{\text{cov}}(X)$ as discussed in Section 6.3 and is deferred to future work.'

Additional Reference

McGaugh, S., Lelli, F. & Schombert, J. 2016, Radial Acceleration Relation in Rotationally Supported Galaxies, Physical Review Letters, 117, 201101.

[This reference was already included in Version 1.0 but is highlighted here for its relevance to Revision 1.]